Missing Element in Sorted Array (medium)

Problem Statement

Given an array `arr` containing `unique` integers in the `ascending` order, and integer `k`, return the kth missing number starting from the `arr[0]`.

The `missing numbers` are those that are not present in the array, and greater than `arr[0]`.

Examples

- **Example 1**:
  - **Input**: `arr = [3, 5, 6, 7], k = 2`
  - **Expected Output**: `8`
  - **Justification**: The missing numbers in sequence are 4 and 8. The 2nd missing number is 8.
- **Example 2**:
  - Input:** `arr = [1, 2, 4, 8], k = 3`
  - **Expected Output**: `6`
  - Justification:** The missing numbers are 3, 5, 6, etc. The 3rd missing number is 6.
- **Example 3**:
  - **Input**: `arr = [2, 3, 4, 7, 8, 10, 11], k = 2`
  - **Expected Output**: `6`
  - **Justification**: The missing numbers in this sequence start from 5, 6, 9... The 2nd missing number in this sequence is 6.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    // Method to find the k-th missing element in a sorted array
    public int findMissingElement(int[] nums, int k) {
        // ToDo: Write Your Code Here.
        return 0;
    }
}
```

Test Case | Run Code Result | Stdout

```
[4,7,9,10]
1
```

Use Example Testcases | Run | Submit

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Solution: Missing Element In Sorted Array

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